

LLMs as Agents in Psycholinguistic Research

Louis Schiekiera (1,2), Max Zimmer (3), Christophe Roux (3), Sebastian Pokutta (3,4),
Fritz Günther (1)

(1) HU Berlin, (2) FU Berlin, (3) Zuse Institute Berlin, (4) TU Berlin

Research Colloquium
Psychological Methods,
WiSe 25/26, HU Berlin

2026-02-05

A (too) brief introduction to LLMs

- LLMs ...
 - ▶ process text in discrete chunks called **tokens**.
 - ▶ consist of a sequence of **layers** (Transformer blocks).
- One layer consists of ...
 - ▶ **Self-attention**: mix info from earlier tokens, making information contextual.
 - ▶ **MLP**: per-token learned **feature update**.
- A token's **hidden state** ...
 - ▶ is its vector representation after each layer
 - ▶ $\approx$ model's evolving "mental" representation

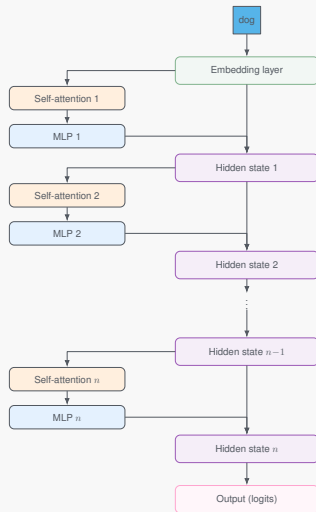


Figure 1: Illustration of a transformer.

- Hidden states are coordinates in a high-dimensional space.
- **Similarity Geometry:** Words with similar meanings should be close together (Mikolov et al., 2013).
- *Intuition:* If the model "knows" that **dog** and **cat** are related, their hidden state vectors should be similar (high cosine similarity).

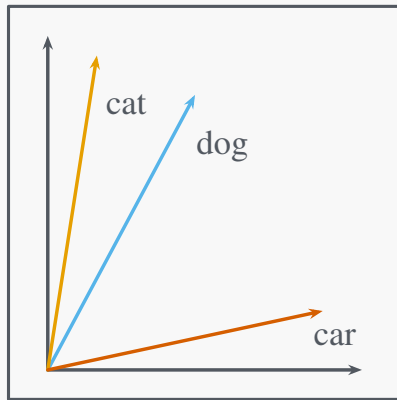


Figure 2: Conceptual illustration of semantic geometry.

Classic Cognitive Science Question

If you only observe a system's (external) **behavior**, how much can you recover about the structure of its (internal) **cognition**? (Baker et al., 2009)

- **Cognition:**

- ▶ What the system "thinks".
- ▶ For LLMs: hidden state representations * (Derby et al., 2021)

- **Behavior:**

- ▶ What the system "does".
- ▶ For LLMs: Generated text (Abramski et al., 2025, e.g., associations, choices, etc.).

* (among other things like self-attention activations, etc.)

Conceptual Approach

Research Question applied to LLMs

If you only observe an LLM's **behavior**, how much can you recover about the structure of its **hidden state** representations?

- **Path A (Internal)**: Extract hidden states
→ Compute similarity matrix $\mathbf{S}^{\text{hid}}$.
- **Path B (External)**: Run behavioral tasks
→ Compute similarity matrix $\mathbf{S}^{\text{beh}}$.
- **Compare**: Do $\mathbf{S}^{\text{hid}}$ and $\mathbf{S}^{\text{beh}}$ look the same?

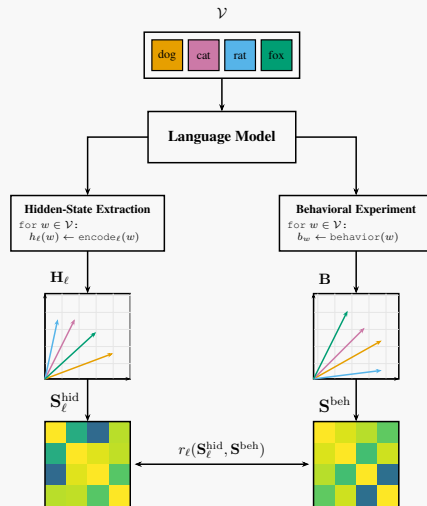


Figure 3: Overview of the RSA approach.

Behavioral Paradigms - Free Association (FA)

Free Association = Open word production

- "What comes to mind?"
- Given 1 input word w_i ; produce 5 output words o_j (De Deyne et al., 2019, 2013; De Deyne and Storms, 2008; De Deyne et al., 2012; Petrenco and Günther, 2025).

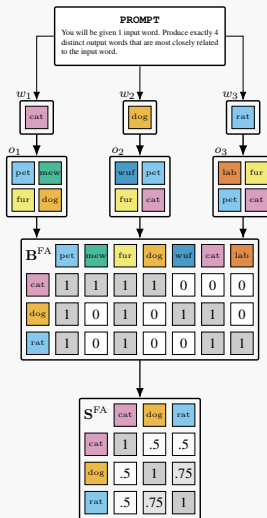


Figure 4: Free Association paradigm.

Behavioral Paradigms - Forced Choice (FC)

Forced Choice = Constrained comparison

- "Which word is most related/similar?"
- Given **1** input word w_i and **16** candidate words c_i , pick the **2** most related/similar output words o_i (Roads and Love, 2021; Demiralp et al., 2014; Li et al., 2016).

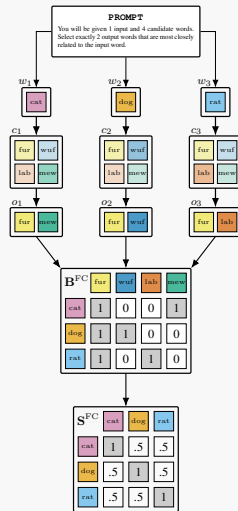


Figure 5: Forced Choice paradigm.

Scale of the Study

- **Models:** 8 Instruction-tuned decoder-only LLMs.
- **Vocabulary:** 5,000 frequent English nouns.
- **Data:** 17.5 Million behavioral trials collected. *
- **Hidden State Extraction:** 4 strategies: Averaged, Meaning, Task (FC), Task (FA).

Table 1: Data collection statistics.

Paradigm	T_{total}	T_{m}	$T_{\text{w+m}}$	A_{t}	$A_{\text{w+m}}$
Forced choice	12.52M	1.565M	313	2	626
Free association	5.04M	0.630M	126	5	630

T_{total} : total number of trials, T_{m} : per model, $T_{\text{w+m}}$: per model and word, A_{t} : Number of associations per trial, $A_{\text{w+m}}$: Number of associations per model and word. Note that $A_{\text{w+m}} = T_{\text{w+m}} * A_{\text{t}}$.

*Thank you HU HPC

Table 2: Model specifications.

Model	n_{params}	n_{layers}	d_{model}
Falcon3-10B-Instruct	10B	40	3072
gemma-2-9b-it	9B	42	3584
Llama-3.1-8B-Instruct	8B	32	4096
Mistral-7B-Instruct-v0.2	7B	32	4096
Mistral-Nemo-Instruct-2407	12B	40	5120
phi-4	14B	40	5120
Qwen2.5-7B-Instruct	7B	28	3584
rnj-1-instruct	8B	32	4096

n_{params} = number of parameters in billions (B); n_{layers} = number of layers; d_{model} = hidden-state dimension width.

Hidden State Extraction Strategies - Meaning

Meaning: Prompt "What is the meaning of..." (Tikhomirova and Wulff, 2026).

$$\mathbf{e}_{\ell}^{\text{meaning}}(w) = h_{\ell}(w, c_{\text{meaning}}(w))$$

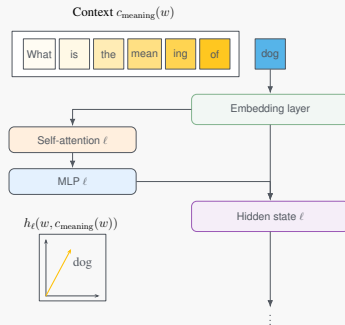


Figure 6: Hidden State Extraction Strategy - Meaning.

Hidden State Extraction Strategies - Averaged

Averaged: Mean over 50 natural sentences from the C4 corpus (Bommasani et al., 2020).

$$\mathbf{e}_\ell^{(\text{avg})}(w) = \frac{1}{50} \sum_{i=1}^{50} h_\ell(w, c_i(w))$$

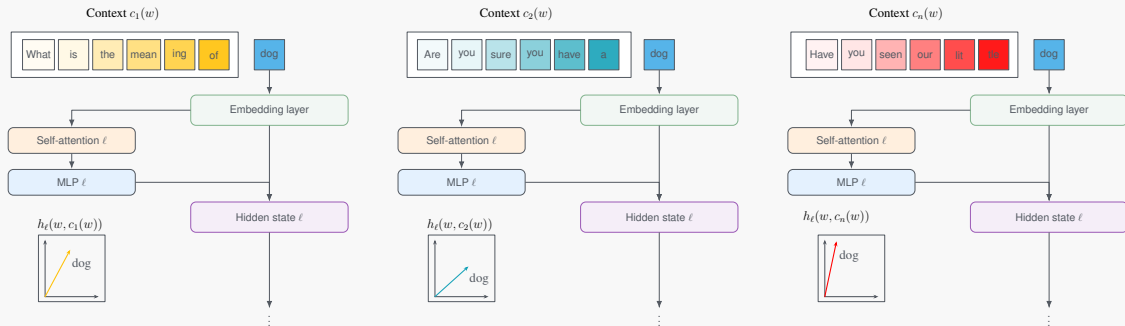


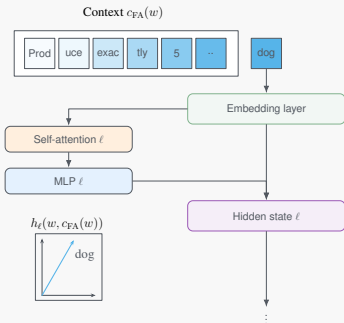
Figure 7: Hidden State Extraction Strategy - Averaged.

Hidden State Extraction Strategies - Task (Forced Choice/Free Association)

Task (FC/FA): Extracted from the exact behavioral prompt.

$$\mathbf{e}_\ell^{\text{task}}(w) = h_\ell(w, c_{\text{task}}(w))$$

Task (Free Association):



Task (Forced Choice):

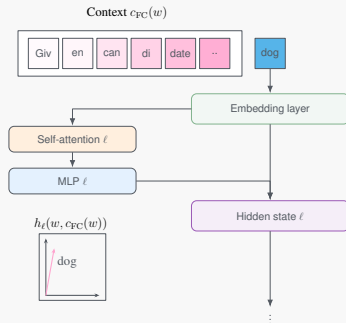


Figure 8: Hidden State Extraction Strategy - Task (Forced Choice/Free Association).

Evaluation (1/3): Representational Similarity Analysis (RSA)

Comparing Representational Spaces

- **Idea:** Compare two representational spaces by correlating their **pairwise similarity structure** (geometry), not coordinates (Kriegeskorte et al., 2008).
- For each model, extraction strategy, and layer ℓ , compute hidden-state similarity $\mathbf{S}_\ell^{\text{hid}}(i, j)$ and compare to a reference geometry $\mathbf{S}^{\text{ref}}(i, j)$.

Reference geometries $\mathbf{S}^{\text{ref}}$

- Behavior: $\mathbf{S}^{\text{FC}}$ (Forced Choice), $\mathbf{S}^{\text{FA}}$ (Free Association)
- Lexical baselines: $\mathbf{S}^{\text{FT}}$ (FastText), $\mathbf{S}^{\text{BERT}}$ (BERT)
- Cross-model consensus: $\mathbf{S}_m^{\text{X}}$ (mean similarity over other models/layers)

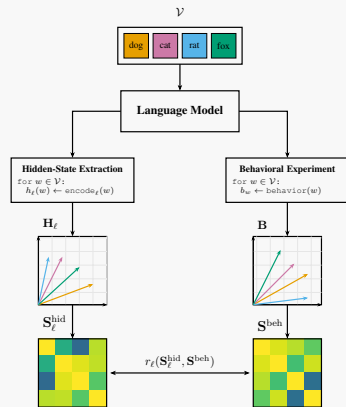


Figure 9: Overview of the RSA approach.

Evaluation (2/3): Nearest-Neighbor Overlap (NN@k)

- **Idea:** A local complement to RSA: do the spaces agree on each word's **nearest semantic neighbors**?
- For a similarity matrix $\mathbf{S}$, define the k -NN index set:

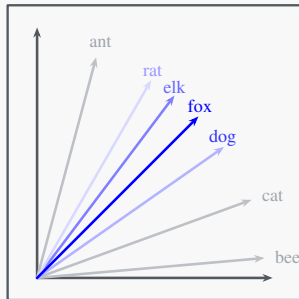
$$N_k^{\mathbf{S}}(i) := \arg \text{topk}_{j \in \{1, \dots, |\mathcal{V}|\} \setminus \{i\}} \mathbf{S}(i, j).$$

$$\text{NN@}k^{(\ell)}(i; \mathbf{S}^{\text{ref}}) := \frac{|N_k^{\mathbf{S}^{\text{hid}}_{\ell}}(i) \cap N_k^{\mathbf{S}^{\text{ref}}}(i)|}{k}.$$

Protocol

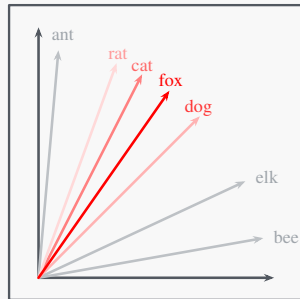
- Evaluate $k \in \{5, 10, 20, 50, 100, 200\}$ against the same references as RSA $(\mathbf{S}^{\text{FC}}, \mathbf{S}^{\text{FA}}, \mathbf{S}^{\text{FT}}, \mathbf{S}^{\text{BERT}}, \mathbf{S}^{\text{X}}_{\text{m}})$.

$$N_3^{\mathbf{S}^{\text{hid}}_{20}}(\text{fox})$$



Hidden-state from phi-4
3 NNs for layer 20

$$N_3^{\mathbf{S}^{\text{FC}}}(\text{fox})$$



Behavior from phi-4
3 NNs for FC paradigm

Evaluation (3/3): Held-out-Words Ridge Regression

- **Goal:** Predict hidden-state similarity for **unseen words** (generalization).
- **Split:** 80% training words / 20% test words; build pairs (i, j) *within* each split.
- **Target:** $y_{ij} = \mathbf{S}_\ell^{\text{hid}}(i, j)$.

$$\mathbf{x}_{ij} := \begin{bmatrix} \mathbf{S}^{\text{FT}}(i, j) \\ \mathbf{S}^{\text{BERT}}(i, j) \\ \mathbf{S}_m^{\text{X}}(i, j) \\ \mathbf{S}^{\text{FC}}(i, j) \\ \mathbf{S}^{\text{FA}}(i, j) \end{bmatrix}.$$

Protocol details

- Fit on $n = 100,000$ sampled training pairs; evaluate on all $n = 499,500$ test pairs.
- report test-set R^2 and incremental gains from adding behavior (FC/FA).

Results: Layer-wise Alignment

- **Consensus:** Different models share a lot of semantic structure ("Universal/Platonic" geometry) (Huh et al., 2024; Kaushik et al., 2025).
- **FC > FA:** FC is more aligned with hidden-state similarity than FA.
- **Baselines:** Bert and FT perform worse than FC.

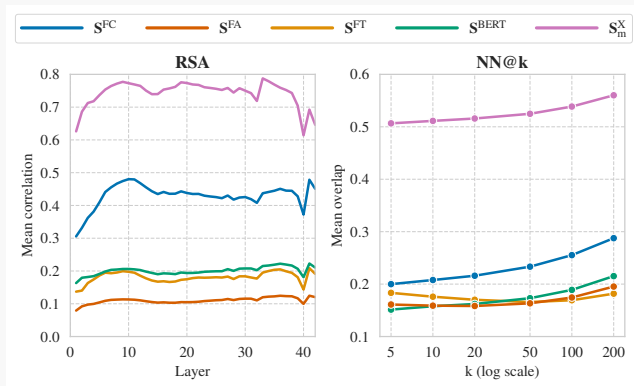


Figure 10: RSA and NN@k for 5 reference geometries.

Results: Layer-wise Alignment

- Task-aligned prompts (Task FC/FA) and Meaning (Meaning) improve alignment in early layers.
- (Averaged) peaks in late layers.

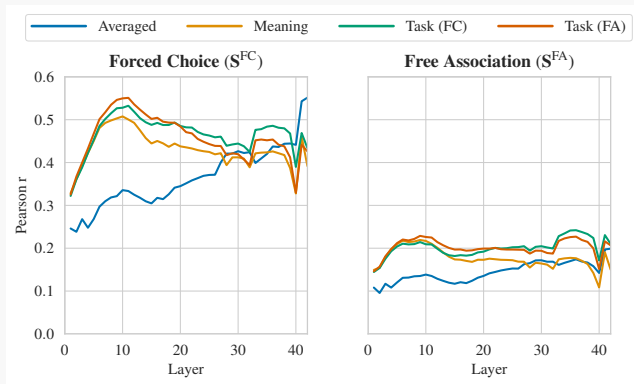


Figure 11: RSA results for FC and FA across 4 extraction strategies.

Regression Results: Can Behavior Predict Hidden States?

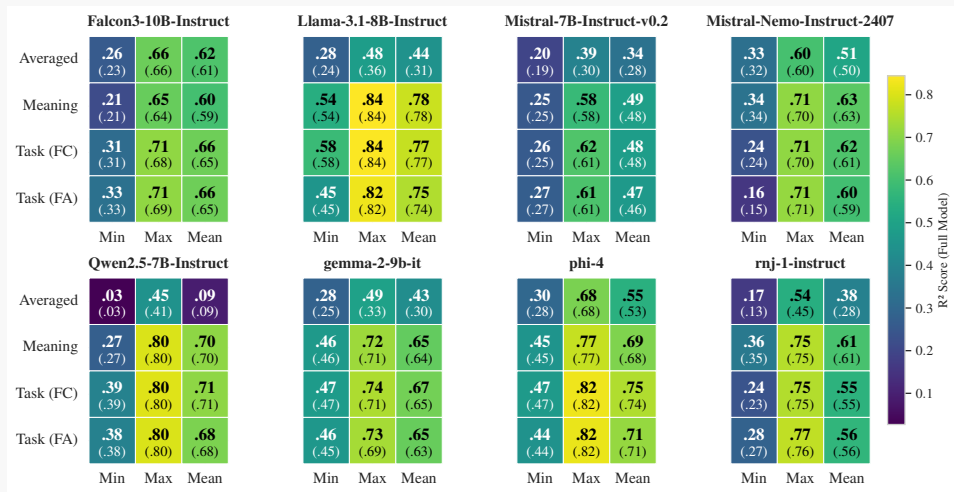


Figure 12: Regression results for FC and FA across 8 models and 4 extraction strategies.

Regression Results: Can Behavior Predict Hidden States?

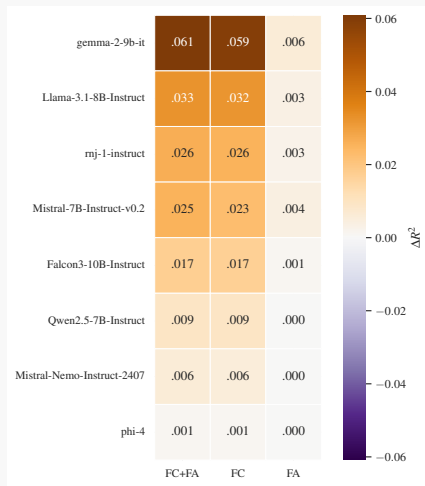


Figure 13: Incremental gains from adding behavior (FC/FA) to the regression model.

The Measurement Insight

Behavior can reveal internal states, but the **measurement instrument** matters.

1. **Conditioning is key:** Constrained tasks (Forced Choice) generate dense, well-conditioned data that recovers geometry well: $\mathbf{B}^{\text{FC}} \in \mathbb{R}^{5000 \times 5000}$.
2. **Open-ended is noisy:** Free Association is too sparse/heavy-tailed to be a reliable map of internal geometry: $\mathbf{B}^{\text{FA}} \in \mathbb{R}^{5000 \times \approx 15000}$.
3. **Implication:** To study the "mind" of an LLM (or human?), use tasks that force explicit comparisons.

- **Scope:** Only English nouns (5k). Do verbs/adjectives behave differently?
- **Prompt sensitivity:** Results depend on the specific prompts and candidate sets used in FC.
- **Correlation $\neq$ Causation:** High alignment doesn't prove the model "uses" this geometry for the behavior.
- **Future:**
 - ▶ Causal interventions (edit the hidden state, see if behavior changes).
 - ▶ Comparison to human behavioral data.

From Associations to Activations: Comparing Behavioral and Hidden-State Semantic Geometry in LLMs

Louis Schiekiera^{1,2} Max Zimmer^{3,4} Christophe Roux^{3,4} Sebastian Pokutta^{3,4} Fritz Günther¹

Abstract

We investigate the extent to which an LLM’s hidden-state geometry can be recovered from its behavior in psycholinguistic experiments. Across eight instruction-tuned transformer models, we run two experimental paradigms—similarity-based forced choice and free association—over a shared 5,000-word vocabulary, collecting 17.5M+ trials to build behavior-based similarity matrices. Using representational similarity analysis, we compare behavioral geometries to layerwise hidden-state similarity and benchmark against FastText, BERT, and cross-model consensus. We find that forced-choice behavior aligns substantially more with hidden-state geometry than free association. In a held-out-words regression, behavioral similarity (especially forced choice) predicts unseen hidden-state similarities beyond lexical baselines and cross-model consensus, indicating that behavior-only measurements retain recoverable information about internal semantic geometry. Finally, we discuss implications for the ability of behavioral tasks to uncover hidden cognitive states.

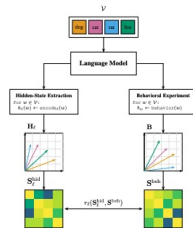


Figure 1. Conceptual overview. For a shared vocabulary V , we (i) extract layer- ℓ word representations to form a hidden-state similarity matrix S_ℓ^{hid} , and (ii) run behavioral association tasks (forced choice/free association) to build a cue-response matrix B and behavioral similarity S^{beh} . RSA correlates the pairwise similarities in S_ℓ^{hid} and S^{beh} to quantify behavior-activation alignment.

Figure 14: Preprint by Schiekiera et al. (2026).

Link: <https://www.arxiv.org/pdf/2602.00628>

References

- Abramski, K., Rossetti, G., and Stella, M. (2025). A word association network methodology for evaluating implicit biases in LLMs compared to humans. *arXiv preprint arXiv:2510.24488*.
- Baker, C. L., Saxe, R., and Tenenbaum, J. B. (2009). Action understanding as inverse planning. *Cognition*, 113(3):329–349.
- Bommasani, R., Davis, K., and Cardie, C. (2020). Interpreting pretrained contextualized representations via reductions to static embeddings. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4758–4781.
- De Deyne, S., Navarro, D., Prefors, A., and Storms, G. (2012). Strong structure in weak semantic similarity: A graph based account. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, volume 34.

- De Deyne, S., Navarro, D. J., Perfors, A., Brysbaert, M., and Storms, G. (2019). The Small World of Words: English word association norms for over 12,000 cue words. *Behavior research methods*, 51(3):987–1006.
- De Deyne, S., Navarro, D. J., and Storms, G. (2013). Better explanations of lexical and semantic cognition using networks derived from continued rather than single-word associations. *Behavior research methods*, 45(2):480–498.
- De Deyne, S. and Storms, G. (2008). Word associations: Norms for 1,424 dutch words in a continuous task. *Behavior research methods*, 40(1):198–205.
- Demiralp, Ç., Bernstein, M. S., and Heer, J. (2014). Learning perceptual kernels for visualization design. *IEEE transactions on visualization and computer graphics*, 20(12):1933–1942.

- Derby, S., Miller, P., and Devereux, B. (2021). Representation and Pre-Activation of Lexical-Semantic Knowledge in Neural Language Models. In Chersoni, E., Hollenstein, N., Jacobs, C., Oseki, Y., Prévot, L., and Santus, E., editors, *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics*, pages 211–221. Association for Computational Linguistics.
- Huh, M., Cheung, B., Wang, T., and Isola, P. (2024). The platonic representation hypothesis. *arXiv preprint arXiv:2405.07987*.
- Kaushik, P., Chaudhari, S., Vaidya, A., Chellappa, R., and Yuille, A. (2025). The Universal Weight Subspace Hypothesis.
- Kriegeskorte, N., Mur, M., and Bandettini, P. A. (2008). Representational similarity analysis-connecting the branches of systems neuroscience. *Frontiers in systems neuroscience*, 2:249.
- Li, L., Song, A., Malave, V., Cottrell, G., and Yu, A. (2016). Extracting human face similarity judgments: Pairs or triplets? *Journal of Vision*, 16(12):719.

- Mikolov, T., Chen, K., Corrado, G., and Dean, J. (2013). Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.
- Petrenco, A. and Günther, F. (2025). Centroid analysis: Inferring concept representations from open-ended word responses. *OSF Preprints*. Version 1.
- Roads, B. D. and Love, B. C. (2021). Enriching Imagenet with human similarity judgments and psychological embeddings. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3547–3557.
- Schiekiera, L., Zimmer, M., Roux, C., Pokutta, S., and Günther, F. (2026). From associations to activations: Comparing behavioral and hidden-state semantic geometry in LLMs. *arXiv preprint arXiv:2602.00628*.
- Tikhomirova, T. and Wulff, D. U. (2026). Where meaning lives: Layer-wise accessibility of psycholinguistic features in encoder and decoder language models. *arXiv preprint arXiv:2601.03798*.